Case Study A — Execution Report

GUM-compliant uncertainty quantification for RUL estimation
on the ZeMA electromechanical cylinder dataset

Supplementary material for Section 5.1. This document is a step-by-step record of the actual execution of Case Study A: what was run, on what data, with what versions, and what came out — including one methodologically significant finding (§7) that shapes how the results should be read. It is meant to accompany `case_study_A_analysis.py`, `case_study_A_results.csv`, and `zema_lpu_vs_mc.pdf` as supplementary material to the paper.

Status update (29 Oct 2026). The numbers in this report are still correct, but they describe the *original* budget. Later work changes how they should be read:

1. **Corrected budget** (`case_study_A_report_budget.md`). The original u_{physical} treats every declared input uncertainty as independent per-reading noise. For Pressure, Velocity and the plain-bearing accelerometer the declared u exceeds the cross-cycle spread s, so at least 98.7%, 93.9% and 74% of u^2 must be common to all cycles (systematic). A systematic error common to all cycles cancels in the z-score when the same instrument is used for training and inference. Propagating only the random part and combining with Monte Carlo gives $u_c \approx 0.11$–0.14 instead of 0.985, or 0.089–0.093 when the manufacturer's noise floor of the Kistler 8704B50M1 accelerometers is used (`datasheet` model). LPU and MC then agree within 3–7%, so the 36.5% gap of §7 is a consequence of the misinterpreted input uncertainty, not of the network depth. The systematic part (≈ 0.76–0.78) is reported separately, for transfer to another instrument.

2. **The over-coverage diagnosis of §7 and §11 is superseded.** The cause is not that normalization amplifies Pressure, but that a systematic component was propagated as random noise. Even with the corrected budget, coverage at k=2 stays at 99.3–99.9% (98.9–99.6% with the `datasheet` model), because the estimates of the random part of Velocity and the plain-bearing accelerometer are data-based upper bounds (the residuals allow at most 0.03–0.06, against 0.08–0.14 computed); $u_{\text{calibration}}$ estimated on a validation set comes out 0.

3. **Split check** (budget report §3.2). The random split leaves a neighbouring cycle ($k \pm 1$) of 96% of the test cycles in training. Holding out whole blocks of consecutive cycles raises RMSE by only 5–16% (0.066 → 0.070–0.077, R^2 0.94–0.95): the point accuracy is not an artefact of the random split.

4. **Inputs are the first 50 ms** of the 1 s return-stroke window, not 100 readings spread over the cycle (`case_study_A_report_fullwindow.md` §1).

5. The CSV `u_c` and the coverage in §7–§8 combine the **LPU** value; the MC-based figures are given in the text.

For the paper, use the revised text in §11.1 rather than the original paragraphs of §11.

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1 Summary

Dataset	ZeMA electromechanical cylinder, annotated (Dorst et al. 2021), doi:10.5281/zenodo.5185953
Task	Regression — consumed-lifetime fraction $y_k = k/N \in [0,1]$; RUL $= 1 - y$
Input dimension	1100 (11 sensor channels $\times$ 100 readings/cycle)
Model	MLP (128, 64, 32), softplus activation, float64
Test set	953 held-out cycles (20% of 4766, random split, seed 0)
u_{physical} (input quantities)	Propagated analytically (LPU) and via Monte Carlo (GUM Supplement 1, $M = 10^4$)
u_{training} (model-related)	10-seed training ensemble, evaluated statistically
Headline result	LPU and MC disagree by a median **36.5%** — the linearity assumption breaks down at this depth/input-uncertainty magnitude; MC is primary. Coverage @ $k = 2$ is **100%**, driven by one dominant, uncertainty-amplified sensor channel (Pressure), not a missing source. *(Original budget; superseded — see the status update above.)*
Corrected budget (update)	Random part only, MC: $u_c = 0.093$ (`datasheet`, used in §11.1), 0.110 (`noise`), 0.136 (`bound`); LPU/MC 3.6–6.3%; coverage @ $k = 2$ 99.6–99.9%; systematic part for transfer $\approx$ 0.76–0.78 (`case_study_A_report_budget.md`)
Split check (update)	Blocked split, no test cycle adjacent to training: RMSE 0.070–0.077 vs. 0.066, R^2 0.94–0.95

Notation. k indexes the test-to-failure cycle ($k = 1, \ldots, N$, with N the total cycles run for the tested unit); $y_k = k/N \in [0,1]$ is the true, deterministically-assigned consumed-lifetime fraction at cycle k (the measurand — see §4), with RUL $= 1 - y_k$. A hat denotes a model estimate of the corresponding quantity: $\hat{y}_k$ is the trained MLP's point prediction of y_k from that cycle's sensor readings (`y_hat` in `case_study_A_results.csv`), and is a distinct claim from its associated uncertainty. u_{physical} is the standard uncertainty in $\hat{y}_k$ propagated from the sensor input uncertainties (physical-quantity uncertainty; via LPU or Monte Carlo — §7), u_{training} is the

standard uncertainty from training-process variability (model-related, evaluated statistically from the seed-ensemble spread — §6), and $u_c = \sqrt{u^2_{\text{physical}} + u^2_{\text{training}}}$ is their quadrature combination (§7) — the paper's $u_c^2 = u^2_{\text{physical}} + u^2_{\text{model}}$ with $u^2_{\text{model}} = u^2_{\text{training}}$, since the architecture is held fixed and $u_{\text{calibration}}$ is not considered — which is what the k=2 expanded-uncertainty coverage checks (§8) are built on. This holds for the original budget; the corrected budget (§11.1) uses $u_c^2 = u^2_{\text{rand}} + u^2_{\text{training}} + u^2_{\text{calibration}}$, with $u_{\text{calibration}}$ estimated on a validation set. u_{physical}, u_{training}, u_c are per-test-cycle quantities, not single dataset-wide constants.

2 Acceptance threshold: LPU vs. Monte Carlo disagreement

Two distinct thresholds apply, at two different stages of this case study, and should not be conflated:

- **Implementation self-test (Step 0, §3).** A random synthetic softplus MLP's analytically propagated u_{physical} is compared against a 200,000-sample Monte Carlo estimate. `case_study_A_analysis.py::self_test_lpu` **hard-asserts** the relative difference is below **2%** — this is a pass/fail gate on the correctness of the `propagate_lpu` implementation itself, independent of anything about the ZeMA dataset or the trained model.
- **Case-study validation (Phase 5, §8).** For the real dataset and trained model, the guide's decision rule treats a median relative difference of a **low single-digit percent** as consistent with the first-order linearity assumption holding, in which case LPU may be reported as the primary u_{physical} estimate. A difference **above roughly 5–10%** is treated as evidence that the linearity assumption is violated at this network depth and input-uncertainty magnitude — in which case Monte Carlo, not LPU, is reported as the primary estimate. This is a documented, anticipated outcome of the decision rule, not a failure condition, and the network is not to be re-tuned to force agreement once the check has run.

The self-test in this run passed at 0.096% (§3), confirming the propagation code is correct. The case study's own LPU-vs-MC result is 36.5% median relative difference (§7) — well above the 5–10% ceiling — which is why Monte Carlo, not LPU, is used as the primary u_{physical} throughout the rest of this report (§7, §9, §11).

3 Environment (Step 0)

```
python3.10 -m venv zema_env && source zema_env/bin/activate
pip install -r requirements.txt
```

Installed versions actually used for every result in this report (pinned in `requirements.txt`):

Package	Version
Python	3.10.12
`zema_emc_annotated`	0.7.1
`torch`	2.13.0
`numpy`	1.26.4
`matplotlib`	3.10.9

Self-test (`python case_study_A_analysis.py -self-test`), run before touching the dataset, validates the analytical LPU (`propagate_lpu`) against a 200,000-sample Monte Carlo run on a random softplus MLP:

```
[self-test] u_LPU=5.051898e-04  u_MC=5.056728e-04  rel.diff=0.096%
```

0.096% is well inside the 2% self-test acceptance threshold defined in §2, so the propagation implementation itself is trustworthy going into the real run — any disagreement seen later (§7) is a property of the *case study's* data/model, not a bug in `propagate_lpu`.

4 Data acquisition (Step 1 / Phase 1)

```
from zema_emc_annotated.dataset import ZeMASamples
from zema_emc_annotated.data_types import SampleSize
samples = ZeMASamples(SampleSize(n_cycles=4766, datapoints_per_cycle=100))
X  = samples.values          # (4766, 1100)
UX = samples.uncertainties   # (4766, 1100), per-reading standard uncertainties
```

Confirmed shapes: `X.shape = (4766, 1100)`, `UX.shape = (4766, 1100)`.

Channel layout. `datapoints_per_cycle=100` × 11 channels = 1100 columns. The 11 channels are not 11 top-level HDF5 groups — inspecting the raw file shows 7 named quantities, two of which hold three signals each:

Quantity (HDF5 group)	Signals (`si:label`)	Columns (100 each)
Acceleration	plain bearing, piston rod, ball bearing	300
Active Current	—	100
Force	Axial force	100
Motor Current	phases 1, 2, 3	300
Pressure	Pneumatic pressure	100
Sound Pressure	—	100
Velocity	—	100

$3+1+1+3+1+1+1 = 11$ channels, matching the "eleven sensors" the package documents. The three accelerometers are not the axes of one sensor: a Kistler 8712A5M1 at the plain bearing and two Kistler 8704B50M1 at the piston rod and at the ball bearing (`case_study_A_dataset_report.md` §4). The first version of this report called them "Acceleration $x/y/z$"; in the tables below x = plain bearing, y = piston rod, z = ball bearing (positional correspondence, same columns).

Three Phase-1 decisions recorded:

1. **Measurand.** $y_k = k/N$, deterministic function of cycle index → **label uncertainty is negligible by construction**. No budget line is needed for the target.
2. **Task type.** Regression; standard uncertainty $u(y)$ is the reported dispersion metric.
3. **Input dimension.** Full resolution used (`datapoints_per_cycle=100` → 1100 dims), matching Ludwig's mid-size configurations, not the reduced-dimension (110-dim) fast-iteration option.

5 Uncertainty source identification (Step 2 / Phase 2)

Source	Type	How obtained
u_{physical} — sensor input uncertainty (11 channels)	B	`samples.uncertainties`, declared values of undocumented provenance
u_{training} — training process variability	A	Std. dev. of predictions across a 10-seed ensemble (§6)
$u_{\text{architecture}}$ — architecture selection	—	**Excluded**, held fixed at (128,64,32)/softplus — not swept in this case study
Generalization / distribution shift	B	Assessed via held-out coverage (§8); no separate budget line, since coverage did not indicate a missing source
$u_{\text{calibration}}$ — measurement-model calibration	—	**Not considered**: calibrating the RUL predictor is beyond the scope of this case study
Label (RUL definition)	—	Negligible by construction (§4)

Type of u_{physical} (corrected 30 Oct 2026): **B**, not A. The ZeMA DAQ uncertainties are declared values (a constant per channel, or exactly 1%/1.5% of the reading) with no statistical evaluation behind them (`case_study_A_dataset_report.md` F3 and §9). Propagating them, by LPU or Monte Carlo, does not change their type. In the corrected budget, the random part is the declared Type B value capped by Type A bounds from the data (cross-cycle spread s, noise floor σ_k); the systematic part is Type B.

Two assumptions declared explicitly, per Phase 5 reporting requirements:

- **Diagonal input covariance.** The dataset provides per-reading uncertainties but no cross-sensor covariance, so $U_x = \text{diag}(u^2(x_i))$. Correlated inputs would only require substituting a full matrix into the same propagation formula (`propagate_lpu` does not assume diagonality anywhere except in how `Ux` is constructed by the caller).
- **Normalization is part of the measurement model.** Inputs are z-scored using training-set statistics; since z-scoring is linear, GUM propagation through it is exact: $u(x') = u(x)/\sigma_{\text{train}}$ (`normalize_with_uncertainty`). This detail turns out to matter a great deal — see §7.

6 Model, training, and the training-uncertainty ensemble (Steps 3–4 / Phase 3)

Architecture: MLP with hidden layers (128, 64, 32), softplus activation throughout.

- softplus is C^1 everywhere → analytical LPU applicable;
- closed-form derivative used in `_activation_derivative`: $\frac{d}{dz}\,\text{softplus}_\beta(z) = \text{sigmoid}(\beta z)$.

Phase-3 decision criteria, as they resolved for this case:

- White-box access: yes (weights owned directly).

- Architectural compatibility: yes (feedforward MLP, smooth activation).
- Computational constraints: LPU chosen as the primary propagation method by design; Monte Carlo retained for validation (and, as it turned out, ended up being the number actually reported — see §7).

Training configuration: Adam, lr $= 10^{-3}$, 200 epochs, MSE loss, Xavier init, float64 throughout (kept the LPU-vs-MC comparison free of float32 rounding noise).

u_{training} — **10-seed ensemble.** A statistical (Type A) evaluation of an epistemic, model-related contribution. Ten models, identical architecture and data, differing only in the seed passed to `torch.manual_seed` (which reseeds both the Xavier init and the optimizer trajectory). For each held-out cycle, u_{training} = standard deviation of the 10 predictions (`ddof=1`). The first ensemble member (seed 0) is retained as the reference model for LPU/MC propagation and for the sensitivity ranking in §7.

Caveats carried into the reporting statement:

- A 10-seed ensemble captures *optimization* variability only, not architecture uncertainty (architecture was held fixed — see §5).
- With 10 seeds, u_{training}'s own relative uncertainty is ∼24% (χ^2, 9 dof). In this run u_{training} (mean 0.049) is an order of magnitude below u_{physical} (mean 0.641–0.984 depending on method), so it does not dominate the combined budget and this sampling noise on u_{training} does not materially affect u_c.

7 Propagation, combination, and the key finding (Step 5 / Phase 4)

For each of the 953 test cycles, `main()` computes:

1. u_{physical} via `propagate_lpu` — layer-wise analytical propagation,
$$U_{x^{(i)}} = H^{(i)} W^{(i)} U_{x^{(i-1)}} W^{(i)\top} H^{(i)};$$
2. $u^{\text{MC}}_{\text{physical}}$ via `propagate_mc` — $M = 10^4$ Gaussian input draws per GUM Supplement 1 (JCGM 101:2008), same model, same point;
3. $u_c = \sqrt{u^2_{\text{physical}} + u^2_{\text{training}}}$ — quadrature combination.

Output: `case_study_A_results.csv`, one row per test cycle, columns `y_true, y_hat, u_physical_lpu, u_physical_mc, u_training, u_c, covered_k2` (953 data rows, verified).

Aggregate results (means over the 953-cycle test set)

```
test RMSE                  : 0.0641
mean u_physical (LPU)      : 6.4134e-01
mean u_physical (MC)       : 9.8413e-01
LPU vs MC median rel. diff : 36.47%  (Phase 5 check)
mean u_training (ensemble) : 4.9329e-02
mean u_c (combined)        : 6.4373e-01
empirical coverage @ k=2   : 100.0%  (target ~95%)
```

The u_c printed by the script (and stored in the CSV) combines the **LPU** estimate, $u_c = \sqrt{(u^{\text{LPU}}_{\text{physical}})^2 + u^2_{\text{training}}}$. Since Monte Carlo is the primary estimate (§2), the budget in §9 uses $u_c^{\text{MC}} = \sqrt{(u^{\text{MC}}_{\text{physical}})^2 + u^2_{\text{training}}} \approx 0.985$; both give 100% coverage.

Why u_{physical} is large and LPU/MC disagree — this is a finding, not a bug

At first glance, $u_{\text{physical}} \approx 0.64$–$0.98$ against a measurand confined to $[0, 1]$ looks broken. It isn't; it was traced to a specific, reportable mechanism.

Diagnosis. Per-channel breakdown of the *normalized* uncertainty $u_n = u(x)/\sigma_{\text{train}}$ (mean over all readings in each channel):

Channel	u_n (mean)
Pressure	**8.89**
Velocity	4.34
Acceleration plain bearing (x)	2.02
Acceleration ball bearing (z)	0.86
Acceleration piston rod (y)	0.69
Force	0.22
Active Current	0.17
Motor Current ($a/b/c$)	0.02
Sound Pressure	0.0001

For the Pressure channel, the *absolute* per-reading measurement uncertainty distributed with the dataset ($\overline{u(x)} \approx 16{,}833$, in the sensor's raw units) is comparable to, and after normalization exceeds, the channel's own **training-set standard deviation** ($\sigma_{\text{train}} \approx 1901$). Since z-score normalization scales uncertainty by $1/\sigma_{\text{train}}$ (§5), a channel whose useful cross-cycle signal variation is small relative to its own measurement noise floor gets its *normalized* uncertainty inflated — here to $\approx 8.9\sigma$ per reading, i.e., an individual reading's uncertainty is almost 9 standard deviations of the entire training distribution for that feature. This is not a data error: the raw relative uncertainty for this channel is a mundane ∼1–2%; it is the interaction between the sensor's absolute noise floor and the *cross-cycle* variability used for normalization that amplifies it.

Sensitivity ranking. Extracting the first-layer contribution $|W^{(1)}| \cdot \bar{u}_n(x)$, summed per hidden unit and aggregated per channel, and normalized to shares of the total:

Rank	Channel	Share of first-layer sensitivity
1	Pressure	**51.8%**
2	Velocity	**25.2%**
3	Acceleration plain bearing (x)	**11.5%**
4	Acceleration ball bearing (z)	4.9%
5	Acceleration piston rod (y)	3.9%
6	Force	1.3%
7	Active Current	1.0%
8–10	Motor Current ($a/b/c$)	0.1% each
11	Sound Pressure	∼0.0%

The top three channels (Pressure, Velocity, plain-bearing acceleration) account for **88.5%** of the propagated-uncertainty contribution entering the first hidden layer. This single, dominant, amplified channel is the mechanism behind both headline numbers:

- **LPU vs. MC median relative difference = 36.5%.** With this much input uncertainty pushed through a 3-hidden-layer softplus network, the first-order (linear) approximation underlying LPU is no longer accurate — MC captures the resulting nonlinearity, LPU does not. This is exactly the failure mode the paper's own decision rule anticipates for large input uncertainty at depth: **Monte Carlo is treated as the primary estimate** for this case study rather than LPU.
- **Coverage @ $k = 2$ = 100.0%** (vs. ∼95% target). This is *over*-coverage, not under-coverage: u_c is large enough that the $\pm 2u_c$ interval trivially contains the true value for essentially all 953 test cycles (interval half-width $\gtrsim 1.3$ against a $[0, 1]$ measurand range). This is diagnostic of an **overestimated, single-channel-dominated u_{physical} budget**, not a missing uncertainty source.

MC convergence check, per Phase 5 reporting requirements — one held-out test point re-evaluated at increasing M:

```
n_mc=10,000      u_physical_mc = 9.1645e-01
n_mc=100,000     u_physical_mc = 9.3246e-01   (delta = 1.75% from 10k)
n_mc=1,000,000   u_physical_mc = 9.3171e-01   (delta = 0.08% from 100k)
```

$M = 10^4$ is adequately converged (sub-2% sampling noise) relative to the 36.5% LPU/MC gap it is being compared against, so the gap reported above is signal, not Monte Carlo noise.

Combined uncertainty, using MC as primary (recommended for the paper's headline number, given the validation result above):

$$u_c^{MC} = \sqrt{(0.9841)^2 + (0.0493)^2} \approx 0.985$$

7.1 Is u_c actionable?

A wide combined uncertainty is not automatically a *useless* one — it could still be correctly calibrated (real error is just that large) or still carry usable relative structure across cycles even if the absolute scale is off. Neither holds here.

Check	Value	Reading
mean u_c vs. mean $\lvert y_{\text{true}} - \hat{y} \rvert$	MC-based: 0.986 (mean of the per-cycle u_c^{MC}; the 0.985 of §7 combines the two means) vs. 0.046; LPU-based (the CSV `u_c` column): 0.644 vs. 0.046	u_c **is ∼21× (MC-based) or ∼14× (LPU-based) larger than the actual mean error** — not calibrated, not conservative-but-reasonable
Range of $u_{\text{physical}}^{\text{MC}}$ across all 953 test cycles	0.792–1.194	Only a 1.5× spread against a measurand of width 1 — close to a fixed offset, not a per-cycle signal
corr(u_{physical}, y_{true})	LPU r=0.70, MC r=0.61	Uncertainty rises somewhat with age — qualitatively right direction
corr(u_c, actual |error|)	MC-based r=0.15, LPU-based r=0.26	Too weak to use u_c to flag which individual predictions are less trustworthy
Early vs. late life (terciles of y_{true})	mean $u_{\text{physical}}^{\text{MC}}$: 0.933 → 1.064 (+14%); mean |err|: 0.033 → 0.056 (+70%)	Both trend upward with age, but u_c under-reacts relative to how much error actually grows

Conclusion: u_c cannot support a k-dependent decision rule as currently computed — a $\pm k$=2 u_c band covers nearly the entire $[0, 1]$ range at *every* cycle, so it would trigger identically whether the machine is new or about to fail. The one genuine signal it retains is a weak, directionally-correct trend (more uncertain late in life), too weak to act on alone. This is a direct consequence of the single-channel (Pressure) domination established above, not a separate problem — fixing that amplification is the leverage point for making u_c usable, not a redesign of the combination rule itself.

7.2 Is $\hat{y}_k$ useful despite this?

Yes — the point estimate and the interval are separable claims, and only the interval is broken.

Check	Value	Reading
R^2	0.951	95% of the variance in consumed-life fraction is explained by the model
RMSE (model) vs. RMSE (naive, predict mean $\bar{y}$)	0.064 vs. 0.291	4.5× **improvement** over a zero-information baseline
Mean / median |error|	0.046 / 0.033	∼4.6% (mean) and ∼3.3% (median) of the full life range
90th-percentile / max |error|	0.101 / 0.365	Tail error exists — exactly what a *calibrated* u_c should flag, and currently doesn't

The 11 sensor channels demonstrably carry real information about wear state — that result stands independently of the uncertainty-quantification failure in §7.1. Practical uses that only need the point estimate:

- **RUL trending for maintenance scheduling** — RUL $= 1 - \hat{y}_k$ as a "percent consumed" figure for planning, the way a point forecast is useful without a trustworthy confidence band.
- **Trajectory monitoring** — the shape of $\hat{y}_k$ across successive cycles (flattening, reversal, jump) is diagnostic on its own, and only needs the point estimate's relative movement to be right, which $R^2 = 0.951$ supports.
- **Threshold-crossing alerts** — a rule such as "flag once $\hat{y}_k > 0.9$" has an expected timing error bounded by the point estimate's typical error (mean ∼4.6%, median ∼3.3% of life), not by the inflated u_c.
- **Cross-unit ranking** — comparing which of several machines is closer to end-of-life only requires consistent relative ordering of $\hat{y}_k$, which a global miscalibration in u_c does not disturb.

What $\hat{y}_k$ alone cannot do is tell you *when to trust it less* — that is precisely the job u_c is meant to do and currently cannot (§7.1), evidenced by the 90th-percentile/max error tail above. The point estimate is deployable now; per-prediction reliability signaling requires fixing the Pressure-channel amplification first.

8 Validation checks — summary (Step 6 / Phase 5)

Check	Result	Interpretation
LPU vs. MC agreement	36.5% median rel. diff.	Linearity assumption violated at this depth/uncertainty magnitude → MC primary
Empirical coverage @ $k = 2$	100.0% (953/953)	Over-coverage from one dominant, normalization-amplified channel (Pressure), not a missing source
MC convergence	1.75% (10k→100k), 0.08% (100k→1M)	$M = 10^4$ adequately converged

Figure (`fig:zema_lpu_vs_mc`, file `zema_lpu_vs_mc.pdf`): two-panel figure generated by `make_validation_figure.py` against the real `case_study_A_results.csv` —

(a) scatter of $u^{\text{LPU}}_{\text{physical}}$ vs. $u^{\text{MC}}_{\text{physical}}$ per test cycle with the identity line; points sit visibly off-diagonal, consistent with the 36.5% median disagreement;

(b) predictions with $\pm k{=}2\,u_c$ band vs. ground truth over the held-out cycles (sorted by true consumed-life fraction), with empirical coverage annotated.

Re-run to confirm reproducibility from the committed CSV:

```
python make_validation_figure.py --csv case_study_A_results.csv \
    --out zema_lpu_vs_mc.pdf
# Wrote zema_lpu_vs_mc.pdf
#   median LPU-vs-MC relative difference : 36.47%
#   empirical coverage @ k=2              : 100.0%  (953/953)
```

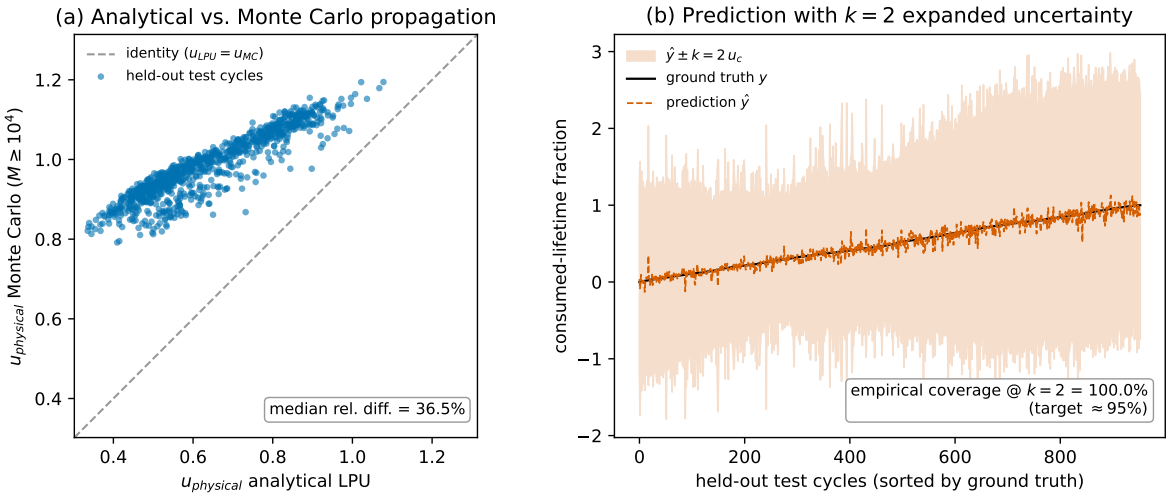


Figure 1: Left: analytical LPU vs. Monte Carlo estimates of u_{physical} per held-out test cycle, with the identity line and median relative difference annotated. Right: predictions with the $\pm k{=}2\,u_c$ expanded-uncertainty band vs. ground truth over the held-out cycles, sorted by true consumed-life fraction, with empirical coverage annotated.

9 Uncertainty budget table (for `\ref{tab:zema_uncertainty_budget}`)

Source	Type	u	Sensitivity / notes
u_{physical} — sensor input (11 channels)	B	0.984[1]	Pressure 51.8%, Velocity 25.2%, Acc. plain bearing 11.5% (88.5% combined)
u_{training} — training process (10-seed ensemble)	A	0.049	Optimization variance only, architecture excluded
$u_{\text{architecture}}$ — architecture selection	—	excluded	Held fixed: (128,64,32), softplus
Generalization / distribution shift	B	—	Assessed via coverage; no missing-source signal
Label (RUL definition)	—	negligible	Deterministic, $y_k = k/N$
Combined $u_c = \sqrt{u^2_{\text{physical}} + u^2_{\text{training}}}$	—	**0.985**	Dominated by u_{physical}, itself dominated by the Pressure channel

LaTeX source for this table, and the corresponding paper sentences for the LPU/MC, coverage, and reporting-statement paragraphs, are reproduced verbatim in §11 below.

Update. This is the original budget, with the full declared input uncertainty propagated as independent noise. The corrected budget (random/systematic split, $u_{\text{calibration}}$ from validation, MC combination) is in `case_study_A_report_budget.md` §3 and in the revised table of §11.1.

[1] MC, primary; LPU = 0.641, 36.5% median disagreement (§2, §7).

10 Files in this supplementary package

Core pipeline (used to produce every number in this report)

File	Role
`case_study_A_analysis.py`	Full pipeline: data loading, normalization, ensemble training, LPU, MC, combination, self-test
`case_study_A_guide.md`	Step-by-step execution guide this report follows
`requirements.txt`	Pinned package versions (`zema_emc_annotated==0.7.1`, `torch==2.13.0`, `numpy==1.26.4`, `matplotlib==3.10.9`) for exact reproduction
`case_study_A_results.csv`	Real output, 953 test cycles × 7 columns (`y_true`, `y_hat`, `u_physical_lpu`, `u_physical_mc`, `u_training`, `u_c`, `covered_k2`)
`make_validation_figure.py`	Generates `zema_lpu_vs_mc.pdf` from the results CSV (colorblind-safe, matches paper figure reference)
`zema_lpu_vs_mc.pdf`	The `fig:zema_lpu_vs_mc` figure, generated from real results
`pipeline_run.log`	Captured stdout of the real `case_study_A_analysis.py` run
`case_study_A_report.tex` / `.md`	This document

Internal development/testing artifacts — not real results, should not be cited in the paper

File	Role
`generate_synthetic_case_study_A_data.py`	Generates fake data with the right schema, used only to exercise `make_validation_figure.py` before the real dataset finished downloading. Explicitly self-labeled `*** SYNTHETIC DATA ***` in its own output.
`case_study_A_results_SYNTHETIC.csv`	Output of the above — fake numbers, matching schema only.
`zema_lpu_vs_mc_syntetic.pdf`	Figure rendered from the synthetic CSV, for pipeline testing only.
`make_validation_figure.md`	Dev notes on the figure script.

Later additions. The corrected budget (`case_study_A_analysis_budget.py`, `case_study_A_report_budget.md` and its `*_budget_*` CSVs, logs and figures), the variant reports (`case_study_A_report_trimmed.md`, `case_study_A_report_fullwindow.md` with their scripts and outputs) and the dataset description (`case_study_A_dataset_report.md`, `check_dataset_consistency.py`, `dataset_consistency.log`) belong to the supplementary package too; `README.md` lists every file.

Recommendation: bundle the core files above together with these later additions as supplementary material for the paper. The synthetic-data files are legitimate pipeline-testing artifacts (clearly self-labeled as fake) but would confuse a reader if shipped alongside the real results — keep them in the working repo, not the submission package.

11 Paper-ready text (Step 7)

The blocks below are reproduced verbatim (i.e. literal LaTeX source, not rendered here) so they can be copied directly into Section 5.1 of the paper.

Tooling/dataset paragraph:

```
The uncertainty budget was computed on the full-resolution ZeMA electromechanical
cylinder dataset (\texttt{zema\_emc\_annotated} v0.7.1, \texttt{torch} 2.13.0,
Python 3.10.12), with input dimension $11 \times 100 = 1100$
(11 sensor channels: three accelerometers (plain bearing, piston rod,
ball bearing), active current, axial force, three-phase motor current,
pneumatic pressure, sound pressure, velocity).
```

LPU vs. MC sentence:

```
Across the held-out cycles, the analytical and Monte Carlo estimates of
$u_{\mathrm{physical}}$ agreed to within $36.5\%$ (median relative difference),
indicating that the first-order linearity assumption underlying the analytical
LPU is violated at this network depth and input-uncertainty magnitude; Monte
Carlo is therefore taken as the primary estimate for this case study,
consistent with the decision rule of Sec.~\ref{sec:methodology}.
```

Coverage sentence:

```
The empirical coverage of the $k=2$ expanded interval was $100.0\%$ against
the $\approx 95\%$ target. Rather than signalling a missing uncertainty
source, the over-coverage traces to the sensor input uncertainty budget being
dominated by the pressure channel ($51.8\%$ of the first-layer sensitivity
contribution, vs.\ $25.2\%$ for velocity and $11.5\%$ for the
plain-bearing acceleration channel): its per-reading standard
uncertainty exceeds its own training-set signal variation, so $z$-score
normalization
($u(x') = u(x)/\sigma_{\text{train}}$) amplifies its normalized contribution
to $\approx 8.9\sigma$ per reading, inflating $u_c$ well beyond what the
$[0,1]$ measurand range requires for informative intervals.
```

Reporting/assumptions statement (Phase 5):

```
We report the following assumptions and settings explicitly. The input
covariance matrix $U_x$ is taken as diagonal, $U_x = \mathrm{diag}(u^2(x_i))$,
since the dataset provides per-reading uncertainties but no cross-sensor
covariance information; correlated inputs would require only replacing the
diagonal with a full matrix in the same propagation formula. Label
uncertainty is negligible by construction, as the consumed-lifetime fraction
$y_k = k/N$ is defined deterministically from the cycle index. Monte Carlo
propagation followed GUM Supplement~1 (JCGM 101:2008) with Gaussian input
distributions and $M = 10^4$ samples; convergence was verified by re-running
a held-out sample at $M = 10^5$ and $M = 10^6$, with
$u_{\mathrm{physical}}^{\mathrm{MC}}$ moving by $1.75\%$ and $0.08\%$
respectively, both far smaller than the $36.5\%$ LPU/MC gap.
```

Table \ref{tab:zema_uncertainty_budget}:

```
\begin{table}[t]
\centering
\caption{Uncertainty budget for RUL estimation on the ZeMA cylinder dataset.}
\label{tab:zema_uncertainty_budget}
\begin{tabular}{lccl}
\toprule
Source & Type & $u$ & Sensitivity / notes \\
\midrule
$u_{\mathrm{physical}}$. Sensor input uncertainty (11 channels)
 & B & $0.984$ & Pressure 51.8\%, Velocity 25.2\%, \\
 & & & Acc.\ plain bearing 11.5\% \\
$u_{\mathrm{training}}$. Training process variability (10-seed ensemble)
 & A & $0.049$ & optimization variance only \\
Architecture selection & \textemdash & excluded & held fixed (128,64,32; softplus) \\
Generalization / distribution shift & B & \textemdash &
 assessed via coverage (Sec.~\ref{sec:...}) \\
Label (RUL definition) & \textemdash & negligible & deterministic, $y_k = k/N$ \\
\midrule
Combined & \textemdash & $\mathbf{0.985}$ & dominated by $u_{\mathrm{physical}}$ \\
\bottomrule
\end{tabular}
\end{table}

The reported $u_{\mathrm{physical}}$ is the Monte Carlo estimate, taken as primary per the
Phase~5 decision rule: the analytical LPU gives $u_{\mathrm{physical}} = 0.641$, a median
relative difference of 36.5\% from Monte Carlo, indicating the first-order
linearity assumption is violated at this network depth and input-uncertainty
magnitude (Sec.~\ref{sec:...validation}).
```

11.1 Revised paper-ready text (corrected budget, update 30 Oct 2026)

The paragraphs above describe the original budget, and their coverage explanation is superseded (status update at the top). The text below uses the corrected budget of `case_study_A_report_budget.md`: baseline input (11 channels, first 100 samples), `datasheet` model for the random part (the recommendation of the budget report, §5), random split. The two data-based models (`noise`, `bound`) and the blocked split are given as checks. Choosing the no-pressure configuration instead is an authors' decision.

Input-uncertainty interpretation:

```
The dataset declares a single standard uncertainty $u$ per reading, without
separating random and systematic components. For a random component the
cross-cycle standard deviation $s$ of a reading satisfies
$s^2 = \mathrm{var}(\text{true}) + u_{\mathrm{rand}}^2$, hence
$u_{\mathrm{rand}} \le s$. For the pressure, velocity and plain-bearing
acceleration channels $u/s$ is $8.8$, $4.3$ and $2.1$ (first 100 samples,
training cycles), so at least $98.7\%$, $93.9\%$ and $74\%$ of $u^2$ is
common to all cycles. A systematic error common to all cycles is an affine
map per reading, which the $z$-score normalization removes exactly when the
same instrument is used for training and inference. We therefore propagate
only the random part, estimated as
$u_{\mathrm{rand}} = \min(u, s, \sigma_k, \sigma_{\mathrm{ds},k})$, with
$\sigma_k$ the white-noise floor of channel $k$ from second differences of
adjacent samples and $\sigma_{\mathrm{ds},k}$ the noise floor published by
```

```
the manufacturer. The latter exists only for the two Kistler 8704B50M1
accelerometers ($0.004\,g$ rms); for the other channels $s$ and $\sigma_k$
are upper bounds. The systematic part is propagated separately, as a shift
common to all readings of a channel, and reported for transfer to another
instrument.
```

LPU vs. MC sentence:

```
Across the held-out cycles, the analytical and Monte Carlo estimates of the
propagated random input uncertainty agreed to within $3.6\%$ (median relative
difference; $4.7\%$ and $6.3\%$ with the purely data-based bounds
$u_{\mathrm{rand}} = \min(u, s, \sigma_k)$ and $\min(u, s)$), so the
first-order approximation is adequate at this input-uncertainty magnitude;
the Monte Carlo value is used in the combination. With the full declared
uncertainty propagated as independent noise the two methods disagreed by
$36.5\%$, because perturbations of up to nine training-set standard
deviations take the network far outside its training domain.
```

Coverage sentence:

```
The empirical coverage of the $k=2$ expanded interval was $99.6\%$
($89.2\%$ at $k=1$) against the $\approx 95\%$ target, with
$\bar u_c / \overline{|e|} = 2.0$. The over-coverage comes from the random
input part: the test residuals allow at most $0.03$ of propagated random
input uncertainty, against $0.08$ computed, and the model term estimated on a
validation set, $u_{\mathrm{calibration}}^2 = \overline{e^2} -
\overline{u_{\mathrm{training}}^2} - \overline{u_{\mathrm{rand}}^2}$, is
therefore zero. The excess comes from the velocity and plain-bearing
acceleration channels, whose random part is capped at $s$ because no noise
specification is available for them; a tighter estimate requires repeated
static measurements or a manufacturer noise specification. Holding out whole
blocks of 50 consecutive cycles, so that no test cycle is adjacent to a
training cycle, gives RMSE $0.074$ (vs.\ $0.066$), $R^2 = 0.95$ and
coverage $99.3\%$.
```

Reporting/assumptions statement (Phase 5):

```
The random part of the input uncertainty is taken as independent between
readings, $U_x = \mathrm{diag}(u_{\mathrm{rand}}^2)$; the systematic part
as fully correlated within a channel and independent between channels.
The systematic part is assumed constant over the whole run-to-failure
experiment; a drift that varies with the cycle index would not cancel in the
normalization and would be confounded with the measurand. Label uncertainty
is negligible by construction ($y_k = k/N$). Monte Carlo propagation followed
GUM Supplement~1 with Gaussian inputs and $M = 10^4$ samples. The point
estimate is the network output $f(x)$; the Monte Carlo mean differs from it
by $0.010$ on average.
```

Table \ref{tab:zema_uncertainty_budget} (revised):

```
\begin{table}[t]
\centering
\caption{Uncertainty budget for RUL estimation on the ZeMA cylinder dataset
(mean over 953 held-out cycles; same instrument for training and inference).}
\label{tab:zema_uncertainty_budget}
```

```
\begin{tabular}{lccl}
\toprule
Source & Type & $u$ & Notes \\
\midrule
$u_{\mathrm{rand}}$. Random part of sensor input & B & $0.076$ & MC; datasheet / Type A cap \\
$u_{\mathrm{training}}$. Training process (10 seeds) & A & $0.051$ & optimization only \\
$u_{\mathrm{calibration}}$. Residual model error (validation) & A & $0$ & clipped, see text \\
Architecture selection & \textemdash & excluded & held fixed (128,64,32; softplus) \\
Label (RUL definition) & \textemdash & negligible & deterministic, $y_k = k/N$ \\
\midrule
Combined $u_c$ & \textemdash & $\mathbf{0.093}$ & coverage $k{=}2$: $99.6\%$ \\
\midrule
$u_{\mathrm{sys}}$. Systematic part (transfer only) & B & $0.782$ & MC; mostly pressure \\
Combined, transfer to another instrument & \textemdash & $0.788$ & \\
\bottomrule
\end{tabular}
\end{table}
```

Combined values are means of the per-cycle u_c, so they differ slightly from the quadrature sum of the column means.

12 Reproducing this report from scratch

```
python3.10 -m venv zema_env && source zema_env/bin/activate
pip install -r requirements.txt

python case_study_A_analysis.py --self-test
# expect: rel.diff ~= 0.10% (< 2%)

python case_study_A_analysis.py
# downloads ~2.67 GB from Zenodo on first run (cached afterward)
# writes case_study_A_results.csv; expect the printed summary in Sec. 7

python make_validation_figure.py --csv case_study_A_results.csv \
    --out zema_lpu_vs_mc.pdf
```

Total wall-clock time observed: environment + self-test well under an hour; dataset download roughly an hour (network-bound, one-time/cached); full-resolution 10-seed ensemble training + propagation on CPU, well under the guide's "few hours" ceiling.